Audit analytics toolkit: Benford, duplicate payments, round sums and journal entry tests

Case study. Sample / demonstration data only. Fiscal year 1 Jan 2025 to 31 Dec 2025.

Problem

Auditors have to look for duplicate payments, threshold avoidance and unusual manual journal entries across populations too large to read line by line. ISA 240 requires testing the appropriateness of journal entries, and the revised ISA 240 applies to audits of periods beginning on or after 15 Dec 2026. This project builds a repeatable set of tests that scores every item and produces a ranked list for targeted follow up.

Data

A generated population for a fictional company: 630 payments to 45 vendors worth 9,499,168.93, and 791 journal entries (1,729 lines). Anomalies were seeded on purpose, and so were benign patterns that look suspicious, such as fixed monthly rent and a fixed licence fee. An answer key records every seeded item, so detection can be measured. Nothing here is real data.

Method

- Benford first digit test on all payments (counts, z statistics, chi square, MAD) and again per vendor, with a minimum of 30 payments.
- Payments: duplicate (vendor, normalised invoice number, amount), near duplicate within 30 days, round sums, just below the 5,000 approval limit, same day splits, and approval below required authority.
- Vendor master: bank account compared with employee bank accounts.
- Journal entries: balance, weekend posting, late posting, year end window, unauthorised user, manual entry to revenue, round totals, missing description, just below the 50,000 limit, manual entry to suspense, number sequence gaps.
- Each flag carries a weight. Items at or above the cutoff score are High. The Excel workbook computes every flag with live formulas, and a separate pandas implementation checks it.

Key findings

Benford, all payments: MAD 0.0218, which is nonconformity on the Nigrini bands, and chi square 54.06 against a critical value of 15.507. Only digits 3 and 4 are significant, and digit 4 is far out (z 6.7). Excluding vendor V031 the MAD falls to 0.0126 (marginally acceptable) and chi square to 11.19. At vendor level V031 is the only vendor flagged (MAD 0.201, 45 payments between 4,700 and 4,990, just under the 5,000 limit).

Test	Flagged	Seeded	Detected	Flagged, not in key
Duplicate payment	10	10	10	0
Near duplicate payment	25	4	4	0
Round sum payment	24	12	12	0
Just below approval limit	55	45	39	4
Same day split payments	34	12	12	10
Approved below authority	3	3	3	0
Vendor fails Benford test	1	1	1	0
Vendor bank equals employee	1	1	1	0
Unbalanced entry	3	3	3	0
Weekend posting	31	8	8	0
Late posting	6	6	6	0
Year end window posting	17	5	5	12
Unauthorised user	5	5	5	0
Manual entry to revenue	5	5	5	0
Round total entry	23	11	11	0
No description	19	9	9	0
Manual just below limit	6	5	5	1
Manual to suspense	4	4	4	0
Sequence gap	6	6	6	0

All 10 duplicate payments were found (76,561.82), including four where the invoice number was reformatted. The employee bank match found vendor V038 (10 payments, 117,824.29). Five manual entries to revenue on 29 to 31 Dec 2025 total 1,310,000.00, all round and two without a description. Three entries are out of balance by 100.00, 75.50 and 1,250.00.

The High list has 49 payments (976,474.76, 10.3% of value) and 8 journal entries. 17 of the High payments are the fixed rent and licence payments, which score High only because of their round or repeated amounts. A reviewer clears them with the contract, so

the tool needs a documented list of recurring vendors before use on real data.

Controls and limitations

A Checks sheet ties payment counts to the Benford and vendor tables, journal lines to entries, debits to credits, and the High lists to the flag columns, and reads PASS. The workbook was recalculated in LibreOffice and compared row by row with the pandas tests: no mismatches in 630 payments and 791 entries, and no formula errors.

Limits: the answer key was written by the same author as the tests, so recall is optimistic and real ledgers are messier. The band test missed 6 of 45 threshold payments that sat below the band, and the vendor level Benford test caught them. Several tests have low precision (year end window: 17 flagged, 5 seeded). Benford does not apply to assigned or fixed amounts. Thresholds and weights are judgment. The output selects specific items for follow up, so it is not an audit sample and results cannot be projected (ISA 530). Real use needs checks on the completeness and accuracy of the data extract first.

What a reviewer should check

- The Checks sheet reads PASS and Parameters match the engagement thresholds.
- The duplicate logic: invoice normalisation ignores spaces, slashes, dots and case, but not transposed digits.
- Trace three High items from the Exceptions list back to source rows and confirm the flags by hand.
- Review the Flagged, not in key column on the Exceptions sheet: those items would need a documented disposition.
- Confirm the Benford result is read with the vendor level table, not on its own.

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